

I Am Multitudes: A Survey and Roadmap for Rethinking Identity in NLP

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Abstract

Natural Language Processing (NLP) systems engage with identity across datasets, models, tasks, and evaluation frameworks, yet they often do so by reducing people to fixed categories (e.g., race, gender, age). This reduction is particularly consequential in identity-sensitive applications: for example, it affects how systems classify harm or generate tailored responses to user needs. But individuals *contain multitudes*: they are not a homogeneous group, but are situated, context-dependent selves whose differences exceed fixed categories. To examine how identity is addressed in NLP, we survey ACL Anthology papers that mention *identity*. Our study investigates how the term is operationalized, where it enters the NLP pipeline, which tasks incorporate identity-related information, and whether its use is explicit or implicit. Our analysis shows that identity is most often operationalized through explicit sociodemographic categories, especially those referring to protected groups by law. Moreover, it is primarily used during the evaluation stage to audit systems for bias and harm. We synthesize these patterns into a taxonomy of identity operationalizations in NLP. Finally, we propose a roadmap to treat identity as relational, contextual, intersectional, and linguistically indexed, with the broader scope of achieving socially aware NLP.

1 Introduction

Users increasingly interact with Natural Language Processing (NLP) systems, including AI chatbots, in high-stakes contexts such as advice-seeking, healthcare, and finance (Chen et al., 2024; Wang et al., 2024). This makes failures especially consequential: recent evidence raises concerns that such chatbots may reinforce delusions, validate self-harm ideation, or provide misleading advice, particularly for vulnerable users (Frances MD and Ramos, 2025). This is partly because appropriate responses depend not only on the content of

a request, but on who is asking: the same question may require a very different answer depending on the user’s identity (Kantharuban et al., 2025; Rodríguez et al., 2025).

However, NLP systems often operationalize identity through demographic proxies, such as gender, race, age, nationality, or other legally protected attributes (Korre et al., 2025; Kwako and Ormerod, 2024). This simplification is limiting because identity is often implied through linguistic cues that shape how the same request should be interpreted (Silverstein, 2003; Bucholtz and Hall, 2005). Concretely, simplified identity representations can result in inappropriate personalization, stereotypical outputs, unequal refusal or helpfulness across identity framings, unsupported inference or optimization toward an abstract default user. For marginalized communities, these harms can further flatten internal diversity and lived experience into exaggerated cultural or linguistic markers (Venkit et al., 2025).

For example, consider the sentence: “*Someone has been threatening me. What should I do?*”. The factual content is that someone is asking for help, regardless of the speakers. However, an appropriate response may depend on the speaker’s lived experience, which may not be explicit in the request itself. This information could be available from prior interaction history or elicited through a follow-up question. For instance, if the speaker has already sought institutional protection but was not taken seriously, simply advising them to contact the police may be insufficient.

This raises a methodological challenge: if identity is not only a sociodemographic category, but also an inferred, context-dependent signal, NLP needs a clearer understanding of how to account for identity. First, we need to answer a central question: **how is identity currently operationalized in NLP, and what forms of identity are included in existing datasets, models, tasks, and evaluation**

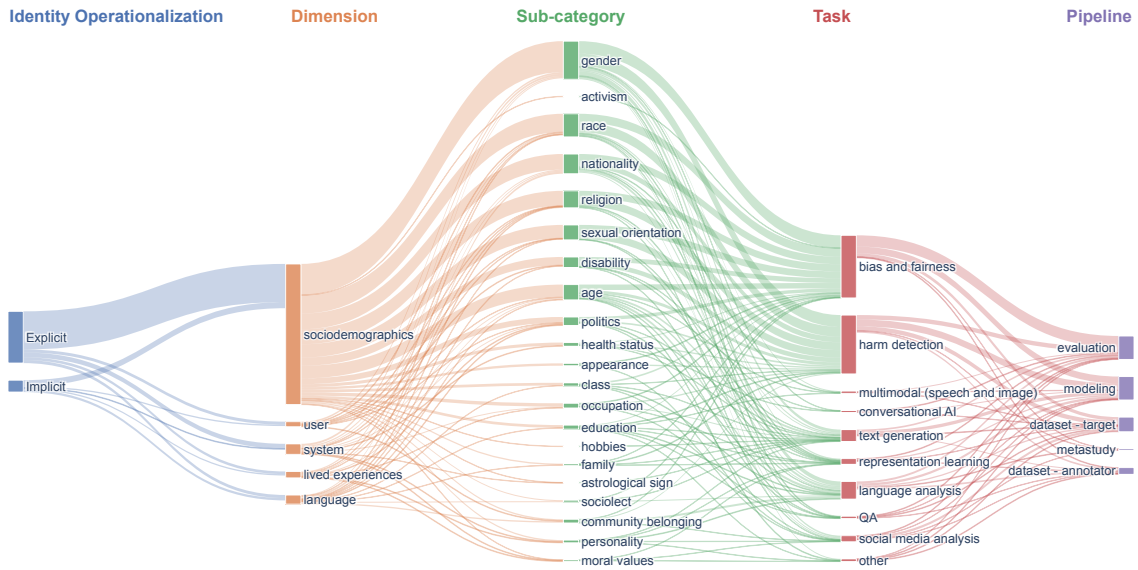


Figure 1: We annotate how identity is operationalized (i.e., explicitly referring to it, or implicitly inferred), which dimensions of identity are considered, what the subcategories are, in what tasks identity is taken into account, and in which stage of the NLP pipeline identity is considered. Each annotation level allows multiple labels.

frameworks? To answer this question, we conduct a systematic review of papers in the ACL Anthology that contain the terms *identity* or *identities*.

We then develop a taxonomy of identity operationalizations and a quantitative measure of identity coverage across the NLP pipeline. The taxonomy captures: (i) the identity dimensions considered, such as sociodemographics, user attributes, linguistic identity, lived experience, or system identity; (ii) the sub-category specific to the identity dimension, such as gender for sociodemographics; (iii) the NLP tasks in which identity is used; (iv) the pipeline stages in which identity is included, such as datasets, modeling, or evaluation; and (v) whether identity is operationalized explicitly (assigned) or implicitly (inferred). Our analysis shows that identity is most often treated as synonymous with sociodemographic categories. It is also concentrated in a limited set of tasks, particularly bias and fairness, and harm detection. This leaves less attention to identity as interactional, inferred and shaped by lived experience. Based on these findings, we propose a roadmap for identity-aware NLP that identifies current limitations and offers concrete recommendations for moving beyond fixed categories toward richer ways of modeling identity.

2 Identity: A Definition

Although there is no consensus on a single definition of *identity* in the social sciences, in NLP we need to define it in order to operationalize it. In sociolinguistics it is widely understood as a relational and socially constructed phenomenon (Bucholtz, 2003; Bucholtz and Hall, 2005). Social identity theory distinguishes between social and personal identity (Turner, 1987). In this framework, social identity refers to the aspects of the self that derive from membership in socially meaningful groups, while personal identity captures attributes that differentiate an individual from others. Social identities are commonly encoded in datasets and annotations, including sociodemographic attributes (e.g., age, gender, race, ethnicity, nationality, disability status, religion, and sexual orientation) and affiliation-based categories (e.g., political alignment). However, while social identity assumes clear group boundaries and internal homogeneity (Holliday, 1999, 2011; Bucholtz, 2003), personal identity encompasses individual-level characteristics that are not reducible to group membership. Such features can be personality traits, personal values, lived experiences, and skills (Holliday, 1999, 2011; Bucholtz, 2003). The combination of both social and personal identity is desirable, and allows identity to be seen as dynamic, context-dependent, and enacted

through language (Holliday, 1999; Block, 2007).

Building on sociolinguistic theories of *indexicality*, identity can emerge through the use of language. The basic idea is that linguistic forms do not only communicate propositional content; they also point to, or *index*, social meanings, stances, and group affiliations (Bucholtz, 2003). This perspective exposes a gap in current NLP systems, where identity is typically operationalized through static demographic attributes. While useful for statistical description, these categories are inherently coarse and treat individuals as members of predefined groups, rather than as socially situated language users (Wang, 2025).

3 Literature Review

To understand how the concept of *identity* is operationalized within NLP research, we conduct a systematic literature review based on the PRISMA guidelines (Page et al., 2021). Through the ACL Anthology API we find all papers whose title and/or abstract contain the keywords *identity* or *identities* published between 2016 and March 2026. We identified 458 unique papers. We expand the pool of papers by adding 12 papers from the 2025 edition of the “Identity-aware AI and Awareness in Learning Agents Workshop”. The annotation taxonomy, including the task categories reported in Table 2, was developed inductively. First, the first author, who has expertise in linguistic approaches to identity, read all abstracts and took exploratory notes on how each paper operationalized identity and the tasks with which it was associated. All three authors then discussed the emerging dimensions using a subsample of 100 papers, grouped overlapping categories, and simplified distinctions where appropriate. The resulting taxonomy and annotation guidelines were agreed upon jointly by all three authors. After the taxonomy was finalized, two authors conducted the remaining annotation, communicating regularly to resolve borderline cases and maintain consistency. Particular attention was given to papers assigned to the *Other* category for tasks or sociodemographic dimensions; these cases were discussed and harmonized throughout the annotation process.

Our annotation scheme captures five aspects of identity use in NLP, allowing multiple labels per paper at each annotation level. Table 1 defines these levels, with guidelines provided in Appendix A. To be included in the final collection, identity must constitute a central component of

the paper: authors must engage with it substantively, and identity must appear in the data, experiments, models, or evaluation. Papers in which identity is only mentioned briefly or used as background motivation are excluded. Specifically, we discard work where identity denotes: (i) identity relations in the context of coreference resolution and information extraction, (ii) authorship attribution (the “identity” of the author), (iii) named entity recognition (the “identity” of an entity), (iv) speaker identification in speech processing, and (v) privacy-preserving methods concerned with masking user identity. After filtering, our final corpus consists of 149 papers. We report in Appendix B the distribution of the papers per year of publication. In Appendix C we report the PRISMA flow diagram of our literature review. Annotated data and code are available at <https://github.com/elisabassignana/survey-identity-nlp>.

4 Identity Dimensions

Figure 1 shows the distribution of the five categories that *identity* can refer to. The following subsections examine these categories in turn, describing how identity is conceptualized in each case and concluding with a short takeaway that summarizes its specific operationalization.

4.1 Sociodemographics

In this dimension, identity is primarily used as a proxy for sociodemographic attributes. This framing aligns with *social identity*, in which social groups are treated as internally coherent and clearly bounded. The largest proportion of papers (84%) falls under this dimension. Most commonly (82%), identity occurs in contexts where sociodemographic groups are examined as targets of discrimination, marginalization, or other forms of harm. These groups are often referred to as protected groups, as they are commonly recognized in legal or ethical frameworks as requiring specific safeguards. Among them, the most prominent categories are gender (22%), race and ethnicity (13%), religion (11%), nationality (11%), sexual orientation (10%), age (9%), and disability (6%). By contrast, health status and class receive comparatively little attention, each accounting for only 2%. The remaining sociodemographic subcategories refer to non-protected groups. These include personality (Iwata et al., 2025; Lu et al., 2025) and one-time occurrences like astrological signs (Sommerauer

Level of annotation	Explanation
Identity Dimension	Identifies what researchers refer to when talking about identity. It distinguishes between system identity, including personas or identities attributed to models, agents, and chatbots; and human identity, including sociodemographics , lived experience , linguistic identity, and user behavioral identity.
ID Subcategory	Identity dimensions are divided into subcategories. We consider the type of demographics such as gender, race, nationality, class, political orientation, personality traits; but also language-related features such as slang, dialect, sociolect, and community identity through language use. Annotators may add additional categories when encountered.
NLP pipeline stage	Records where identity is operationalized in the NLP pipeline, including datasets in terms of targeted entities and annotators, modeling or experimental design, evaluation , and meta-level studies such as surveys or position papers.
Task	Annotates the NLP task in which identity is operationalized, analyzed, or evaluated. The fixed list of tasks, along with a definition, can be found in Appendix D. However, annotators may introduce additional task labels when a paper does not fit existing categories.
Operationalization type	Distinguishes explicit operationalizations, where identity is provided through labels, metadata, or prompts, from implicit operationalizations, where identity is inferred from linguistic or contextual signals.

Table 1: Annotation levels for analyzing how identity is operationalized in NLP papers.

et al., 2025), feminism and LGBT activism (Hede et al., 2021) and hobbies (Tint et al., 2026).

Identity operationalization: *fixed sociodemographic categories.*

4.2 User

In this dimension, identity refers to the behavioral and interactional patterns associated with users in online environments. Tasks include user profiling (Liang et al., 2021), personalization (Kantharuban et al., 2025; Vijjini et al., 2025) or the analysis of posting patterns in online communities. For instance, Holur et al. (2022) analyze social media discourse to determine whether a user is an insider or an outsider of an online conspiracy community.

Identity operationalization: *online behavioral patterns associated with users.*

4.3 Linguistic identity

Linguistic identity captures cases in which identity is operationalized through patterns of language use that index membership in a social group. These works typically treat identity as something inferred from a speaker’s linguistic repertoire. Identity as language includes dialect features (Lucy et al., 2024), code-switching practices (Bouamor et al., 2025; Liang et al., 2025), and stylistic or pragmatic choices (Lu et al., 2025). These features are often used as proxies for group membership, as, for example, community-specific slang can signal affiliation (Sun and Xu, 2022). At the same time, such features can be strategically used to obscure

meaning for outsiders. A salient case is the use of dogwhistles (Dénigot and Burnett, 2020): terms or expressions that appear innocuous in general usage but carry derogatory meanings within a particular community. Conversely, Yee et al. (2023) examine how slurs can be strategically reclaimed by marginalized groups as markers of identity.

Identity operationalization: *linguistic repertoire.*

4.4 Systems

Systems identity captures cases in which identity refers to artificial agents and the personas they adopt or are assigned. Papers explore how conversational agents simulate (Sommerauer et al., 2025), or are perceived as (Ghanbari Haez and Dragoni, 2025) having identities, including demographic traits (Kumar et al., 2024; Rampazzo et al., 2025), personality (Lee et al., 2025), or character roles (Lu et al., 2024). In a small number of cases, the systems dimension incorporates lived experience through biographical or experiential information, such as individuals’ daily routines and preferences collected through open-ended essays (Lee et al., 2025), biographies (Li et al., 2024), and characters’ backgrounds and prior conversations (Lu et al., 2024). Identity is therefore conceptualized either as an assigned feature (e.g., persona-based prompting) or as an emergent property of model behavior.

Identity operationalization: *the constructed persona of an artificial agent.*

4.5 Lived Experience

Work on discourse and narrative identity shows that people construct who they are by telling stories about what has happened to them, how they position themselves, and how they relate those experiences to broader social categories (Bamberg et al., 2011). This dimension grounds identity in personal histories, biographical anecdotes, and socially meaningful experiences. Unlike sociodemographic categories, lived experiences are not always directly observable and cannot be easily reduced to stable group labels. They capture the within-group differences that shape social identity. They are often expressed through open-ended essays (Lee et al., 2025), interviews (Awad et al., 2025) and surveys (Mor-Lan et al., 2025).

Identity operationalization: *narratives grounded in personal and social experience.*

5 NLP Tasks

Table 2 shows the distribution of NLP tasks. Identity is most frequently operationalized within classification-based paradigms, especially in the case of bias and fairness, and harm detection.

Among bias and fairness tasks, we identify different lines of research: assessing biases in language models (Lin et al., 2022; Liu et al., 2024; Zulaika and Saralegi, 2025; Chakour and Sadat, 2025; Devinney et al., 2024); mitigation strategies (Cheng et al., 2021; Panda et al., 2022; Zhang et al., 2020); personalization (Vijjini et al., 2025) and alignment (Lerner et al., 2024). Closely related, harm detection is very present, reflecting the prominence of hate speech (Yoder et al., 2022; Kim et al., 2023), toxicity (Chvasta et al., 2022; Van Dorpe et al., 2023) and abusive language (Lommel et al., 2025) detection in NLP research. Other areas include the evaluation of counterspeech with respect to sociodemographics (Chung and Bright, 2024; Kiritchenko et al., 2025); mental health issues (Liang et al., 2021; Nakajima Wickham, 2023) and conspiracy theories (Holur et al., 2022). In language analysis tasks, identity is relevant when language is used to signal community membership (Sun and Xu, 2022; Aggarwal et al., 2023; Holgate et al., 2018) or attitude shifts within specific populations (CH-Wang and Jurgens, 2021).

In text generation tasks, identity is mostly present in the simulation of characters (Lamia et al., 2025; Lu et al., 2025) or individuals (Sommerauer et al., 2025) and in narrative generation for an

Task	Count
Bias and Fairness	54
Harm Detection	41
Language Analysis	23
Text Generation	13
Social Media Analysis	11
Representation Learning	6
Conversational AI	5
Image and Speech Processing	5
Question Answering	4
Other	9

Table 2: Distribution of NLP tasks. A paper annotated with multiple tasks contributes once to each task.

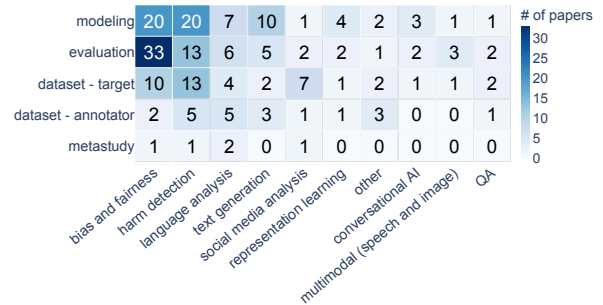


Figure 2: Co-occurrences of pipeline stage where identity is operationalized and NLP tasks.

identity-specific audience (Yunusov et al., 2024). Text generation intersects with harm detection in the production of counterspeech (Chung and Bright, 2024) and with bias and fairness in generating responses according to stereotypical cues (Neplenbroek et al., 2025). Identity-related stereotypes are also at the center of image processing and generation tasks (Ghosh and Caliskan, 2023; Ungless et al., 2023; Jha et al., 2024). Among social media analysis, identity is examined through the analysis of emoji usage (Li et al., 2020) or political discourse (Mor-Lan et al., 2025). A smaller subset investigates identity within language modeling and representation learning, examining how pretrained models encode demographic signals, especially in word embeddings (Smirnov, 2024; Lepori, 2020).

6 NLP Pipeline

Figure 1 shows that identity-related research is mainly operationalized in the evaluation (39%) and modeling (38%) stages of the NLP pipeline, followed by the creation of datasets (34%). Within the latter, we consider two distinct levels of identity dimensions: as properties of the target being annotated, and as dimensions of annotator features. Only 3% of the papers are meta-studies, i.e., surveys and position papers. Task categories tend to

cluster around specific parts of the NLP pipeline. To examine this pattern, we compute co-occurrence matrices between task categories and the pipeline stages in which identity is operationalized. Figure 2 reports the resulting distribution of co-occurrences.

6.1 Datasets

For what concerns annotator features, gender (Pei and Jurgens, 2023; Chen et al., 2025; Biester et al., 2022; Gromann et al., 2023) and race (Pei and Jurgens, 2023; Larimore et al., 2021; Yunusov et al., 2024) are the most prominent. Susanto et al. (2025) consider also education and occupation, while Shem-Tov and Rabinovich (2025) consider how annotators’ personality relates to their musical preferences. Among targets, Mitchell et al. (2025) create a dataset for stereotypes on gender, nationality, age, race, sexual orientation, disability, class, health status, occupation, and politics, being one of the most complete work concerning sociodemographics. Holgate et al. (2018) creates a dataset that considers identity as both sociodemographics and language, studying the interaction between different subcategories (gender, age, class, education, religion, politics) and vulgar expressions.

6.2 Modeling

Most of the modeling pipeline co-occur with the task of bias and fairness (13%) and harm detection (13%) (see Figure 2.) In both settings, identity is generally represented as a fixed and predefined attribute, which is either injected into the model through persona prompting (Draetta et al., 2025) or as input labels for classification tasks (Yoder et al., 2022). The common ground of these experiments is that all individuals are treated as part of a homogeneous group. For instance, Das et al. (2025) analyze shifts in response to personas with varying political orientations (left vs. right), religious identities (Hindu vs. Muslim), and national affiliations (Bangladeshi vs. Indian), ignoring individual differences among these groups. Another prominent case of identity in modeling is related to debiasing techniques (Zhou et al., 2021; Sturman et al., 2024) or bias mitigation (Attanasio et al., 2022; Panda et al., 2022).

6.3 Evaluation

Identity is used as an evaluation lens. Datasets are assessed according to whether their design choices lead to performance disparities across different identities (Santy et al., 2023; Knupleš et al.,

2024); models are tested to determine whether their predictions, generations, explanations, or representations change depending on the target identity (Sweeney and Najafian, 2019; Zhang et al., 2020; Lepori, 2020; Balkir et al., 2022). In bias-related work, identity categories are inserted into controlled contexts to measure whether models associate groups with stereotypes (Sommerauer et al., 2025) or unequal outcomes (Ghanbari Haez and Dragoni, 2025). Similar counterfactual designs appear in work on harm detection and machine translation, where identity terms, either explicit or implicit through names (Ghanbari Haez and Dragoni, 2025), pronouns (Lauscher et al., 2023) or dialectal features (Tatman, 2017), are varied while the surrounding context is kept stable.

A second line of work evaluates representational bias in model internals or outputs, for example, through word embeddings (Sweeney and Najafian, 2019; Smirnov, 2024) and completion analysis of language models (Nozza et al., 2022). Models’ refusals are addressed in Hoehn et al. (2025), where they compare the LLM response according to the presence versus the absence of neurodivergence self-disclosures in prompts.

In vision tasks, identity is evaluated in relation to stereotypical representations of non-normative identities, as in the case of non-cisgender individuals (Ungless et al., 2023). Only a smaller subset of papers moves toward evaluating whether systems can respond appropriately to identity as it emerges through language (Aggarwal et al., 2023; Lucy et al., 2024). They operationalize identity using language features like stance markers.

6.4 Meta-study

Metastudies contain position papers and surveys. Among position papers, work such as Oh (2025) points out that moderation standards are not universally neutral, they privilege some social positions over others; Lin (2025) show how current NLP systems often optimize for an abstract average user while overlooking differences in social position, vulnerability, experience, and communicative norms; Pawar et al. (2025) argue that identity-aware LLMs require cultural reasoning. Among surveys, Niti and Prawati (2024) studies how language is used as an identity marker in Chinese communities; Lauscher et al. (2022) provides an overview of third-person pronoun issues in NLP, especially how current systems fail to handle neopronouns.

7 Operationalization: Explicit vs Implicit

Identity in NLP can be operationalized either explicitly or implicitly. Explicit operationalizations rely on directly stated identity categories, such as sociodemographic labels, metadata, or persona prompts (e.g., “you are a Black woman”), where identity is predefined and externally assigned. By contrast, implicit operationalizations infer identity from indirect signals, such as linguistic style, lexical choices, or statistical associations learned from language. An implicit operationalization of identity reflects the sociolinguistic assumption that language indexes social position, lived experience, and group membership. They are particularly important because they reveal how NLP systems can infer, encode, or ascribe identity even in the absence of explicit sociodemographic information.

Implicit instantiations of identity include:

Identity inferred through language. An example is [Jagfeld et al. \(2021\)](#), which infers gender from the language used on Reddit, and [Pawar et al. \(2025\)](#), which infers nationality by prompting LLMs with names. This includes work on word embeddings, where identity is encoded statistically in patterns of language use. An example is [Lepori \(2020\)](#) for analyzing intersectional biases against Black women.

Indirect ways to address targets. An example is [Parent et al. \(2025\)](#), where they study the stereotypical language used to target groups, such as *brown shirts* for ANTIFA and *blue-haired pronoun-havers* for liberals; [Ghanbari Haez and Dragoni \(2025\)](#) insert identity-marked names into the same scenario in order to test whether models produce systematically different responses depending on the perceived identity of the individual involved.

8 Roadmap for Identity-Aware NLP

Our analysis reveals limitations in how identity is currently operationalized in NLP. We summarize these limitations, and we propose a roadmap to address them through concrete recommendations. The roadmap spans from identity definition to dataset design, modeling, evaluation, and alignment. Across these stages, simplified operationalizations can impose coarse categories during data collection, treat group members homogeneously during modeling, produce stereotypical or unequal behavior during evaluation, and encourage caricatural representations or adaptation toward a default user during alignment.

8.1 Conceptual Ambiguity

The term *identity* is used inconsistently, often interchangeably with sociodemographic attributes, especially legally protected attributes. This ambiguity is not merely terminological: it determines which differences an analysis can detect and which conclusions it can support. For example, if a study operationalizes identity through gender or racial labels, an observed “identity effect” is an effect associated with those predefined categories; the analysis cannot establish whether the effect also depends on lived experience, linguistic identity, context, or differences among people assigned to the same category. Treating such labels as identity in its entirety therefore normalizes a reductionist account and may lead researchers to generalize category-level findings to heterogeneous individuals.

Recommendation 1

Ground the definition of identity in theoretical work, by stating which theory of identity is adopted.

A positive example is [Lee et al. \(2025\)](#), who provide a multidimensional definition of identity drawn from the sociolinguistic concept of social and personal identity. They distinguish between social identity, personal identity, and personal life context. This decomposition makes it possible to test what each component contributes.

8.2 Reductionism

During data collection and annotation, identity is frequently reduced to binary or coarse-grained categories (e.g., male/female, liberal/conservative), ignoring fluidity, intersectionality, and heterogeneity among members of a group, oversimplifying social reality. This occurs because identity is operationalized as a fixed category to make datasets compatible with traditional supervised classification frameworks. With LLMs, however, both inputs and outputs can accommodate richer, self-described, and context-sensitive identity representations.

Recommendation 2

Collect self-identification descriptions rather than imposing fixed categories.

In practice, researchers can complement structured demographic fields with open-ended self-

descriptions, interaction histories, or questions that allow participants to indicate which aspects of identity are relevant in a particular setting. These representations should not be collapsed into fixed labels, and participants should be able to revise them as their circumstances change. This allows people to define which aspects of their identity are salient in a given context. For example, [Weeber et al. \(2026\)](#) prompt LLMs with users’ past conversation histories for better personalization. However, conversation histories may contain sensitive information that users did not intend to disclose or reuse for identity modeling. Their use should therefore require informed consent, data minimization, purpose limitation, and user control over which interactions are retained and used.

8.3 Static Modeling

A core limitation of current NLP approaches is that they often treat identity in isolation from the communicative situation in which language is produced. Yet, identity shapes meaning through the interplay of speaker and receiver characteristics, social relations, context, social norms, culture and ideology, and communicative goals ([Hovy and Yang, 2021](#)). When identity is considered in isolation, these dimensions are flattened, obscuring how identity shifts across receivers, situations, and time.

Identity should be located not only in the speaker/author but also in the receiver/evaluator. As concerns time, longitudinal studies are needed to account for the fact that identity is not stable over time but may change alongside users’ attitudes, experiences, and social positioning.

Recommendation 3

Incorporate interactional variables and perform longitudinal analysis.

For instance, [Chen et al. \(2020\)](#), [August et al. \(2020\)](#) and [Occhipinti et al. \(2025\)](#) consider the receiver perspective. For what concerns longitudinal analysis, [Abercrombie et al. \(2025\)](#) shows the importance of tracking consistency and change across time, suggesting that identity-sensitive NLP should model temporal variation rather than treating users as fixed entities.

8.4 Evaluation

A key limitation of current evaluation frameworks is that they primarily test whether models behave differently toward predefined sociodemographic

groups, while overlooking how identity functions contextually. We propose five complementary evaluation lenses for identity-aware NLP.

First, **subjective task adequacy** should test whether a response accounts for identity-relevant constraints rather than assuming a default user. For example, advice to “contact the police” may be inadequate when prior context indicates that the user has already sought institutional protection without receiving support.

Second, **refusal and safety evaluation** should use matched prompts that preserve the request while varying only explicit or implicit identity cues. Researchers can then compare refusal rates, helpfulness, and the reasons provided to determine whether safety policies are applied asymmetrically.

Third, **identity representation in data** should document which identities, perspectives, and social positions occur in task-specific and training data, including their intersections and within-group variation. This should extend beyond sociodemographics because models often reflect WEIRD norms ([Santy et al., 2023](#); [Mihalcea et al., 2025](#); [Sen et al., 2025](#)).

Fourth, **underspecification** should test whether a model recognizes when identity-relevant information is missing. Rather than rewarding an answer based on unsupported assumptions, evaluation should test whether the model asks a targeted clarification question, such as whether the user has access to a suggested service.

Finally, evaluation should move **beyond output-level measures** by incorporating user-reported outcomes. Models should be evaluated for how interactions affect users’ self-reported well-being across identities.

Together, these lenses extend beyond personalization: they assess not only whether a response is adapted to a user, but whether the system identifies relevant context, manages missing information, avoids unsupported assumptions, and produces appropriate outcomes.

Recommendation 4

Evaluate whether models handle identity context-sensitively across tasks, data, uncertainty, and user outcomes.

Recent work on underspecified reasoning has begun to evaluate whether models recognize when a prompt lacks necessary information, rather than forcing them to produce an answer. For example, [Li](#)

et al. (2025) introduces QuestBench to test whether LLMs ask appropriate clarification questions under underspecification. Identity-aware evaluation can extend this direction by asking whether models manage uncertainty around identity specifically. Luettgau et al. (2026) measure whether AI advice affects users’ well-being before, immediately after, and weeks after interaction.

8.5 Alignment

A central direction for identity-aware NLP is pluralistic alignment: systems should account for diverse values, needs, and perspectives rather than optimizing for a single default user (Sorensen et al., 2024). In the context of identity, pluralistic alignment requires context-sensitive adaptation: systems should use identity information only when it is relevant, justified, and beneficial for the task. The goal is neither maximal personalization nor identity-blindness, but the appropriate use of identity in context. Such adaptation must avoid reducing users to caricatures or stereotypes. A well-aligned system should know when identity information matters, when it does not, and when uncertainty requires clarification.

Recommendation 5

Support pluralistic alignment through context-sensitive use of identity information.

A useful positive example is Halpern et al. (2026), who model diverse human preferences as a distribution over reward functions rather than collapsing disagreement into a single majority preference. At the same time, context-sensitive identity use must avoid user reduction. Cheng et al. (2023) show that LLM simulations of sociodemographic personas can become caricatural: instead of producing a response that is meaningfully grounded in the topic, models may over-emphasize identity-related features and generate flattened, homogeneous depictions of a group.

9 Conclusions

We examined how identity is currently operationalized in NLP and identified recurring issues, focusing on the reduction of identity to fixed sociodemographic categories. Our findings indicate that identity is often treated as an external auditing variable: something used to test whether systems behave differently across groups, rather than something integrated into how language data are collected,

how meaning is represented, or how models are designed. We argue that these practices overlook the relational, indexical, and context-dependent nature of identity as understood in sociolinguistics. To address these gaps, we propose a roadmap emphasizing limitations and concrete recommendations on how to integrate identity throughout the pipeline.

Limitations

First, our corpus includes only papers whose title or abstract contains the terms identity or identities. While this makes the review systematic and reproducible, it may exclude work that engages with identity-related phenomena but does not explicitly use this terminology in the title or abstract. Conversely, some papers may use the term identity in the title or abstract but only engage with it marginally or in a sense that differs from our definition; we address this through manual screening, but some boundary cases remain. Second, our review is limited to the ACL Anthology. This focus allows us to characterize how identity is operationalized within the NLP community, but it may exclude relevant NLP-related work published in other venues, including interdisciplinary venues where identity, language technologies, and social context are also studied. Third, our annotation is conducted at the paper level. This enables broad coverage, but it may limit our ability to capture variation within individual papers, especially when identity is operationalized differently across datasets, models, or evaluation settings. Fourth, our taxonomy simplifies a complex theoretical construct. Although all annotation levels are multilabel, allowing papers to be assigned to multiple dimensions, the categories we identify can still overlap in practice. Our review therefore describes how identity is currently operationalized in NLP, without claiming to provide a comprehensive theory of identity as a social phenomenon. Finally, interactional and longitudinal approaches introduce practical challenges. Modeling combinations of speaker, receiver, social relation, and context increases experimental complexity and may produce sparse or confounded comparisons. Longitudinal data collection is costly, requires sustained participation, and may not be feasible for all projects. Alternatives include repeated cross-sectional evaluation, targeted robustness tests that vary selected contextual factors, or the use of existing longitudinal data collected with appropriate consent. Researchers should state which

dimensions they cannot capture rather than treating a static representation as comprehensive.

Ethical Considerations

The roadmap is intended as a research agenda rather than a requirement to collect or infer increasingly detailed identity information. Richer representations introduce privacy and consent risks: self-descriptions and conversation histories may reveal sensitive information that is irrelevant to the current task, may be reused outside their original purpose, or may enable systems to infer identity attributes that users did not intend to disclose. Researchers should therefore follow data-minimization and purpose-limitation principles, obtain informed consent, and give users control over which information is retained and used. When persistent conversation histories are unnecessary, systems should instead elicit the minimum task-relevant context through targeted clarification.

Clarification and adaptation have their costs. Repeated questions may burden users or solicit sensitive information, while excessive adaptation may over-ascribe identity and reproduce stereotypes. Systems should therefore request identity-related information only when it is necessary to improve the relevance, safety, or appropriateness of the response. The goal is calibrated identity awareness, not maximal personalization or identity inference.

The authors used AI tools, specifically ChatGPT, in order to check grammar, spelling and paraphrasing. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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A Annotation Guidelines

Annotation Guidelines

Annotate each paper at the **paper level** rather than at the sentence level. Base the annotation on:

- the main contribution of the paper;
- the identity concept that is explicitly studied or implicitly assumed.

If multiple categories apply, select all relevant labels.

General notes

- Mark papers in which *identity* is close to the concept defined in the paper.
- Mark papers in which identity is not used in a binary or static way, but is instead self-described, fluid, intersectional, or otherwise goes beyond binarism and static identity categories.
- Exclude shared tasks.

1. Identity Dimension

Annotate whether identity refers primarily to a **human identity** or to a **system identity**. Select the subcategory as well. Subcategories may apply to both humans and systems.

a) Sociodemographics

Examples include gender, race, class, education, disability, and health.

Annotate:

- which sociodemographic dimensions are included.

b) Lived Experience

Identity as narrative identity, backstory, or personal history.

Examples:

- survivor of violence;
- immigrant experience.

c) Linguistic Identity

Identity expressed through language use.

Examples:

- dialects, such as African American English;
- style, register, code-switching, sociollect;
- dogwhistles;
- community identity expressed through language use.

d) User Behavioral Identity

Identity inferred from or represented through user behavior.

Examples:

- interaction patterns;
- preferences;
- user profiles;
- online behavior.

e) System Identity

Identity attributed to the model itself or to simulated agents, AI systems, or chatbots.

Examples:

- model persona;
- chatbot identity;
- simulated agent identity.

2. Pipeline Stage

Goal: identify where identity is operationalized in the NLP pipeline.

a) Datasets

Identity is present in the data, for example through:

- annotators, meaning who labels the data;
- targets, meaning who or what is described.

b) Modeling and Experiments

Identity is operationalized during modeling or experimental design.

Examples:

- pre-training;
- fine-tuning or post-training;
- alignment, such as RLHF or safety layers;
- prompting, such as persona prompting.

c) Evaluation

Identity is used in metrics, benchmarks, qualitative analysis, evaluation frameworks, or stress-testing settings, but is not necessarily part of the modeling stage.

Examples:

- studying bias in word embeddings or datasets;
- linguistic analysis with respect to identities;
- studying the effect of gender on interpreting;
- studying bias in toxicity classification with respect to identity groups;
- proposing identity-related metrics or evaluation frameworks.

d) Meta-study

The paper is a survey, position paper, taxonomy, or other meta-level contribution.

Language

Specify the languages studied in the paper.

3. Tasks

Goal: identify the NLP task to which identity is applied.

a) Harm Detection

Examples:

- hate speech, toxicity, or offensive language;
- misinformation;
- mental health.

b) Emotion / Sentiment Analysis

Identity is applied to emotion detection, sentiment analysis, or affective language understanding.

c) Language Analysis

Examples:

- style, pragmatics, or discourse;
- lexical or morphological analysis;
- story analysis.

d) Question Answering

Identity is applied to question answering tasks or QA systems.

e) Conversational AI

Examples:

- dialogues;
- interactions;
- multi-turn conversation.

f) Bias and Fairness

Identity is used to study, measure, mitigate, or evaluate bias and fairness.

g) Machine Translation

Identity is applied to machine translation or interpreting-related tasks.

h) Social Media Analysis

Identity is applied to social media content, users, communities, or platforms.

i) Other

Use this category if none of the above fit. This label requires a free-text note specifying the task.

4. Identity Operationalization Strategy

Goal: identify how identity is represented in the paper.

a) Explicit

Identity is directly provided or assigned.
Examples:

- identity labels, such as “female” or “Black”;
- metadata, such as a Twitter bio;
- persona prompts.

b) Implicit

Identity is embedded in language use or context and can be inferred from signals such as:

- linguistic features;
- style, tone, or framing;
- contextual signals;
- clusters of annotations used to infer annotator identity;
- language patterns associated with a particular group or condition.

chine translation, text summarization, paraphrase generation, and story generation.

2. Harm Detection Tasks in which the focus is on protected groups. This includes hate speech and toxicity detection, misinformation, mental health.

3. Representation Learning Research on learning general-purpose representations of language. Examples include masked language modeling and word embeddings.

4. Bias and Fairness Research explicitly investigating representational harms, unintended bias and fairness in NLP systems. This includes debiasing methods, bias evaluation frameworks, and fairness-aware modeling.

5. Language Analysis Tasks include error analysis of classifiers, the analysis of style, pragmatics, discourse; lexical or morphological analysis; story analysis.

6. Social Media Analysis This includes analysis of user-generated content on online platforms such as Twitter and Reddit, outside explicitly harmful settings.

7. Conversational AI Tasks involving interactive dialogue systems, where models engage in multi-turn conversations with users. Examples include chatbots, virtual assistants, therapeutic agents, and role-playing systems.

8. Question Answering Tasks where a model answers a specific question, typically within a single-turn interaction and with the goal of providing a factual, informative, or contextually correct answer. Unlike conversational AI, these tasks usually do not require maintaining dialogue context or interactional coherence across multiple turns. A

9. Speech and Image This includes tasks in which the main outcome is speech processing or image generation and processing.

10. Other Any other tasks encountered.

B Paper Distribution

We report in Figure 3 the distribution of the papers included in our literature review per year. Note that 2026 stops in March.

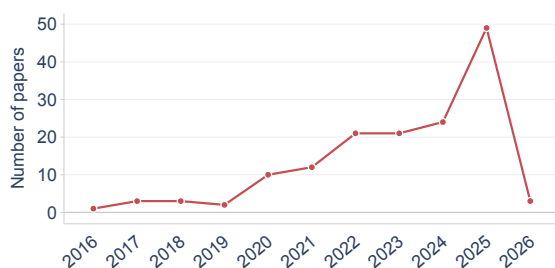


Figure 3: Paper distribution per year.

C PRISMA

We report in Figure 4 the PRIMA flow diagram of the literature review.

D Tasks

1. Text Generation Tasks in which a system produces natural language output. This includes ma-

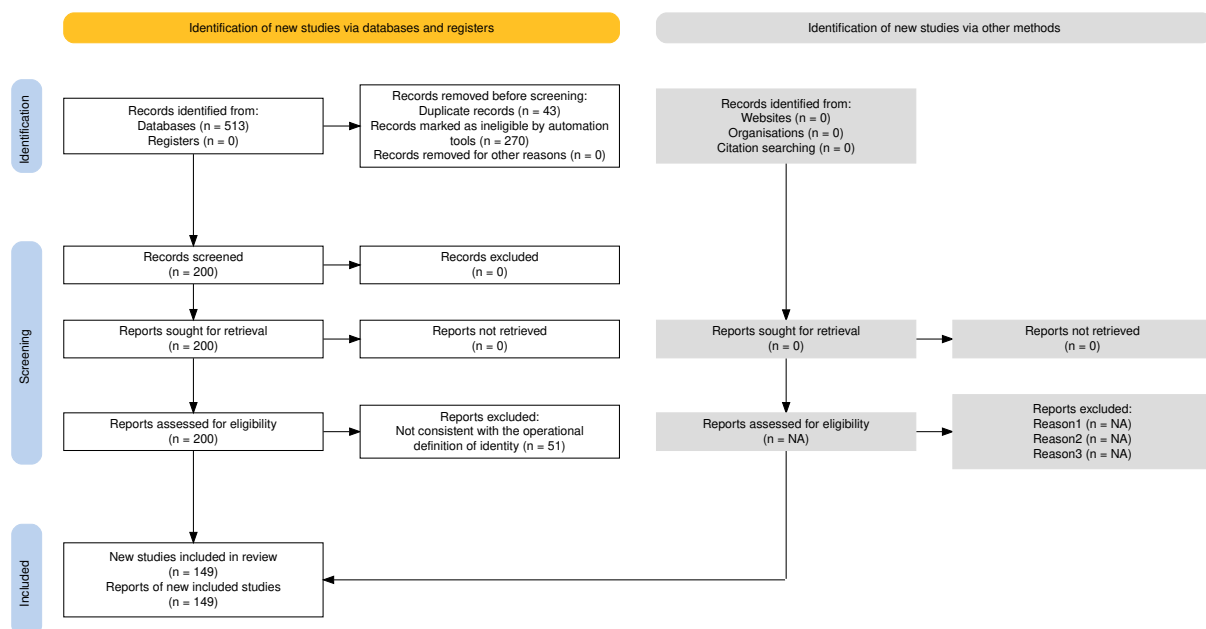


Figure 4: PRISMA flow diagram of our paper collection.